

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

SUBJECT: Generative AI

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Case Study – 5

"DocuMind : AI Document Summarization and Question Answering System"

***Subject**: Generative AI*

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Objective

In modern education and research, efficiently analyzing lengthy documents is essential for improving productivity, simplifying information retrieval, and supporting effective learning. Traditional methods of reading and searching documents manually can be time-consuming and may make it difficult to identify important information, understand key concepts, and prepare study materials.

To address these challenges, this project develops an AI Document Summarization and Question Answering System called DocuMind AI that combines PDF text extraction, information retrieval, prompt engineering, and large language model (LLM) capabilities. The system extracts text from uploaded PDF documents, divides the content into searchable chunks, and uses TF-IDF-based retrieval and the Google Gemini API to generate document summaries, answer user questions, and extract key points. It also generates multiple-choice questions and important exam-preparation questions with answer explanations. Page references are included to help users verify generated answers against the original document. An interactive Streamlit dashboard provides document upload, processing, summarization, question answering, study-material generation, and downloadable outputs.

Therefore, this project focuses on:

- Developing an AI-powered document analysis system for PDF summarization and question answering.
- Extracting and preprocessing text from uploaded PDF documents while preserving page numbers.
- Retrieving relevant document passages using TF-IDF-based information retrieval.
- Generating concise and detailed summaries, key points, and important terminology using an LLM.
- Answering document-related questions with supporting page references and generating MCQs and exam-preparation questions.
- Providing an interactive Streamlit dashboard for document processing, analysis, and downloading generated outputs.

Problem Statement

In modern education and research, efficiently analyzing lengthy documents is essential for improving productivity, simplifying information retrieval, and supporting effective learning. However, traditional methods of document analysis primarily rely on manual reading and searching, which can make it difficult to identify important information, understand key concepts, and prepare study materials.

Traditional document-processing approaches may also struggle with lengthy PDFs, scattered information, and locating relevant passages for specific questions. General-purpose language models may generate answers that are not adequately supported by the source document. Therefore, there is a need for an intelligent document analysis system that combines PDF text extraction, information retrieval, and large language models to generate reliable, document-based summaries and answers.

Therefore, the problem addressed in this project is to develop an AI Document Summarization and Question Answering System that can:

- Extract and preprocess text from uploaded PDF documents while preserving page references.
- Generate concise and detailed summaries of document content.
- Retrieve relevant passages to answer user questions based on the uploaded document.
- Extract important points, key concepts, and terminology from documents.
- Generate multiple-choice questions and important exam-preparation questions with answer explanations.
- Provide an interactive dashboard for document analysis, question answering, and downloading generated outputs.

Implementation Setup

Three main types of document-processing tasks were considered:

- PDF text extraction and preprocessing
- Document summarization and question answering
- Study-material generation, including MCQs and important questions

Model Types

The application uses a large language model (LLM) with retrieval-based document processing. The following components are implemented:

- TF-IDF Retrieval: Identifies document passages relevant to a user's question.
- Google Gemini LLM: Generates summaries, answers, key points, and study questions using the retrieved text.
- Prompt Engineering: Guides the model to answer using the supplied document excerpts and provide page references.

Architecture

The application follows this workflow:

PDF Upload	Text Extraction	Chunking + Page Index	TF-IDF Retrieval	Gemini LLM Output
Input	PyMuPDF	Overlapping text chunks	Relevant passages	Summary / Q&A / Study tools

The generated outputs include document summaries, question-answering responses, key points, MCQs, and important exam-preparation questions.

Technology Stack

Technology	Purpose
Python	Core application logic and text processing.
Streamlit	Interactive web interface, tabs, forms, upload controls, and downloads.
PyMuPDF	Extracts text from PDFs page by page.
Scikit-learn TF-IDF	Ranks document chunks by lexical relevance to a user query.
Google Gemini API	Generates summaries, answers, key points, MCQs, and important questions.
python-dotenv	Loads API configuration from a local .env file.

Core Modules

- PDF processing: reads the uploaded PDF and extracts text from each page, preserving page numbers.
- Chunking: splits page text into overlapping chunks so retrieval can focus on relevant passages.
- Retrieval: uses TF-IDF vectorization and cosine similarity to select chunks related to a user question.
- Prompt construction: instructs the LLM to use supplied excerpts, cite available page numbers, and state when evidence is insufficient.
- Generation: calls the Gemini API to produce summaries, answers, key points, MCQs, and important questions.
- Interface and export: Streamlit tabs expose the features and allow users to download generated text.

Implementation Details

- PDF text is extracted with PyMuPDF, with page numbering retained from the original document.
- Text is normalized by removing null characters, repeated spaces, and excessive blank lines.
- Chunks are created with a maximum character length and overlap to preserve local context.
- For question answering, the system retrieves up to five relevant chunks and sends them with the question to the configured Gemini model.
- For long documents, the current summary implementation selects representative chunks rather than passing every chunk to the model in one request.
- Users can choose summary length, MCQ count, and difficulty level, and can generate important exam questions.
- The app supports downloads of summaries, key points, Q&A history, MCQs, important questions, and extracted text.

Observations / Results

The developed DocuMind AI application provides PDF upload and text extraction, document summarization, question answering, key-point extraction, study-material generation, and downloadable outputs.

Model and Feature Evaluation Results

Feature	Input / Configuration	Observed Result
PDF text extraction	`DL.pdf`	Successful
Pages processed	15 pages	15 pages with extracted text
Text extraction	PDF content	25,094 characters
Searchable chunks	TF-IDF-based chunk retrieval	15 chunks
Document summarization	Standard summary	Generated
Key-point extraction	Document content	Generated
Question answering	Question about confusion matrices	Answer with supporting page references
MCQ generation	Selected number and difficulty	Questions with options and explanations
Important questions	Document content	Questions with answer outlines

Text export	Extracted document	Downloadable `.txt` file

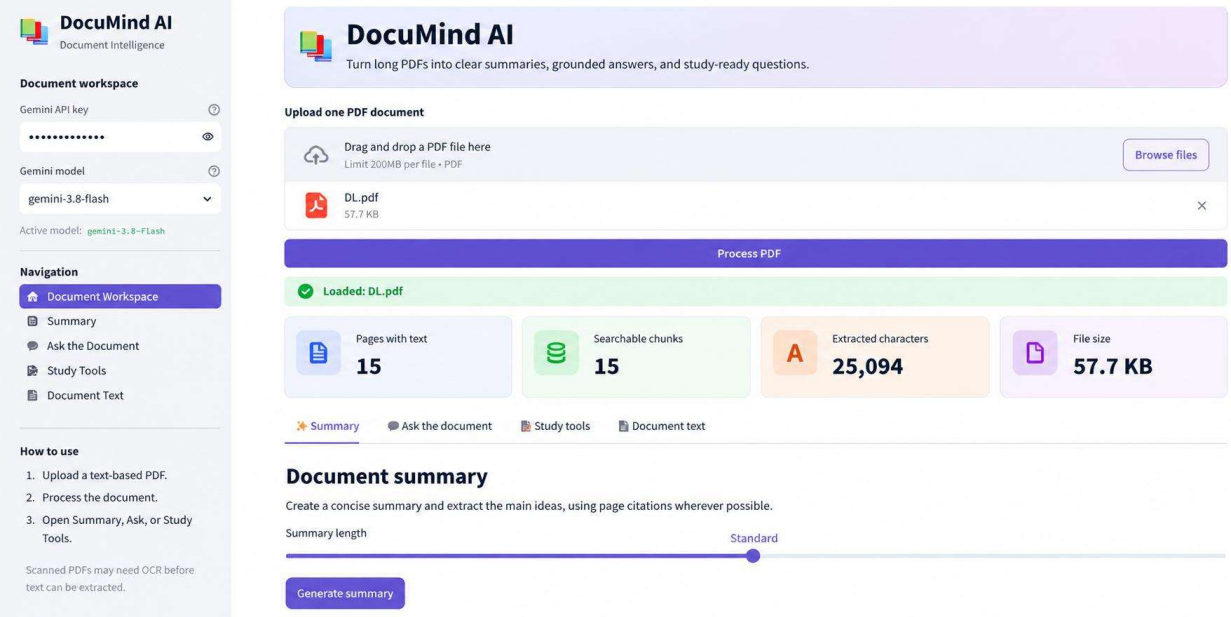
Key Observations

- The application successfully extracted text from the uploaded 15-page PDF.
- The summarization module generated a document overview, important points, and terminology.
- The question-answering module retrieved relevant document passages and generated an answer with page references.
- The study tools generated multiple-choice questions with answer explanations and important questions with answer outlines.
- The interface allowed users to inspect extracted text page by page and download generated outputs.
- The current TF-IDF retrieval method provides a lightweight way to identify relevant passages, but it may miss information when a question uses wording different from the source document.
- Long-document summaries use representative chunks, which means some sections may not be covered in the generated summary.

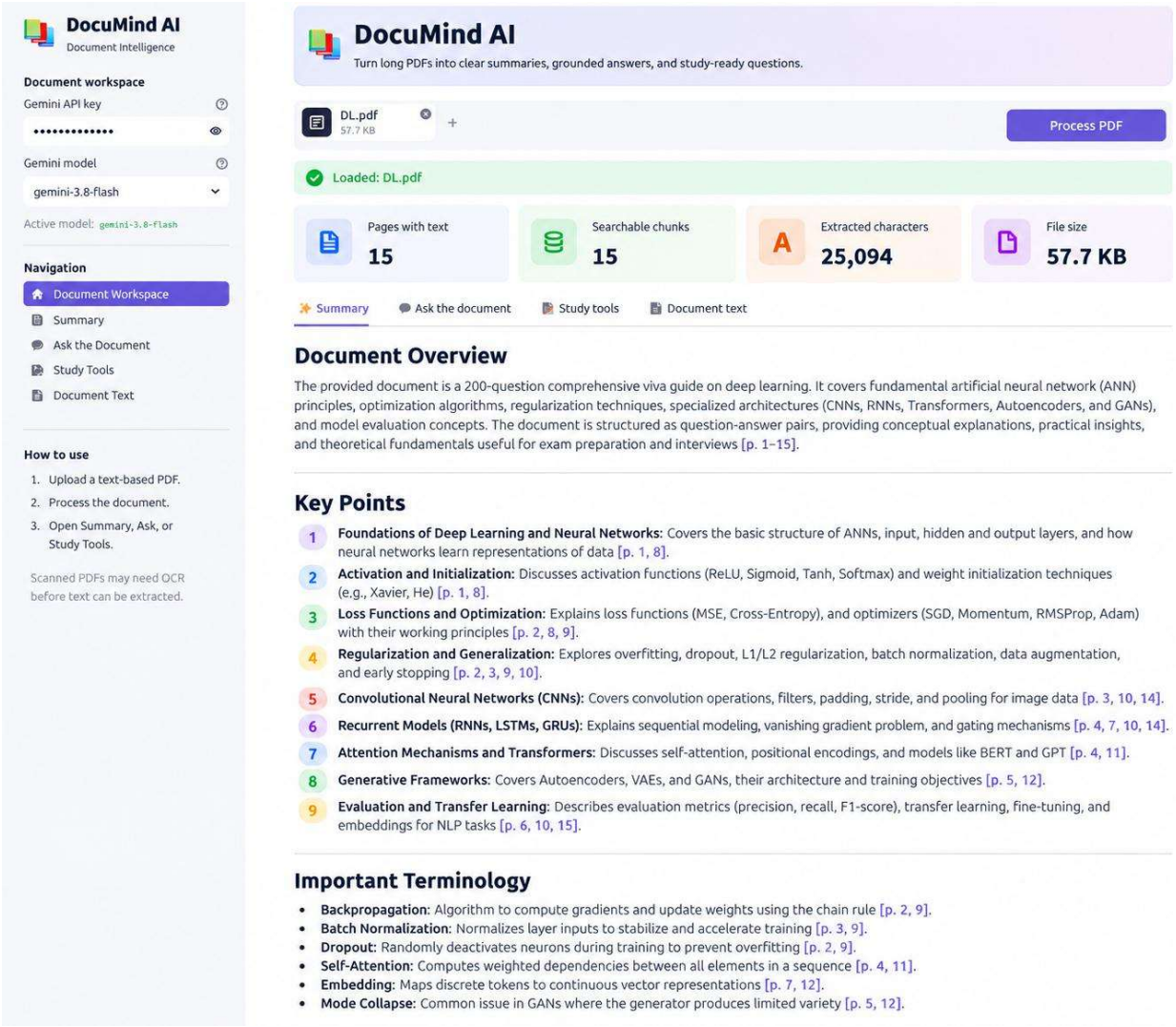
Screenshots / Outputs

The implementation produced:

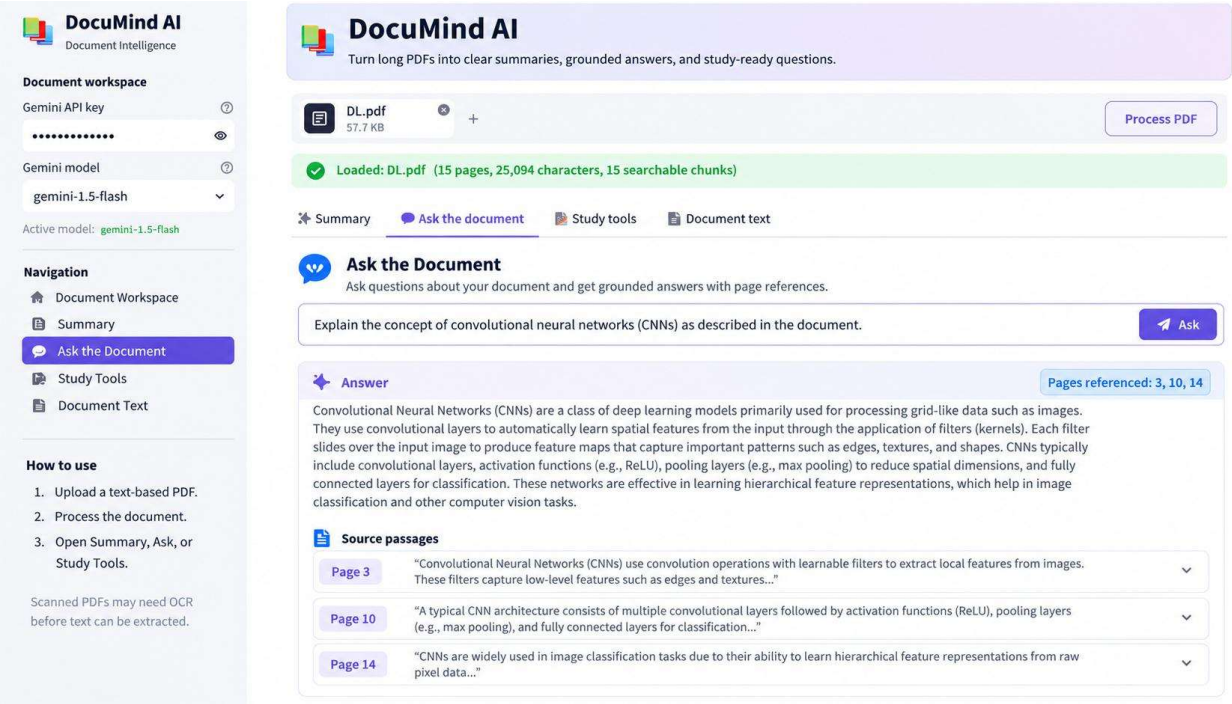
- Document processing dashboard — Displays the uploaded PDF, extracted page count, searchable chunks, and character count.



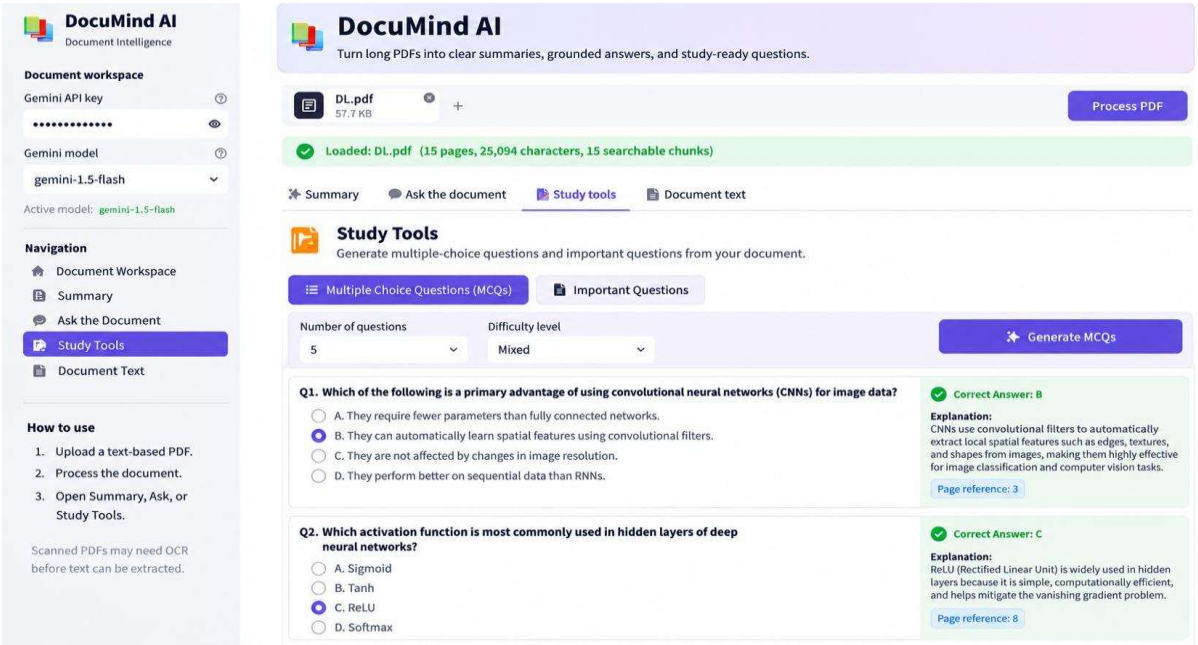
- Document summary — Displays the generated overview, key points, and important terminology.



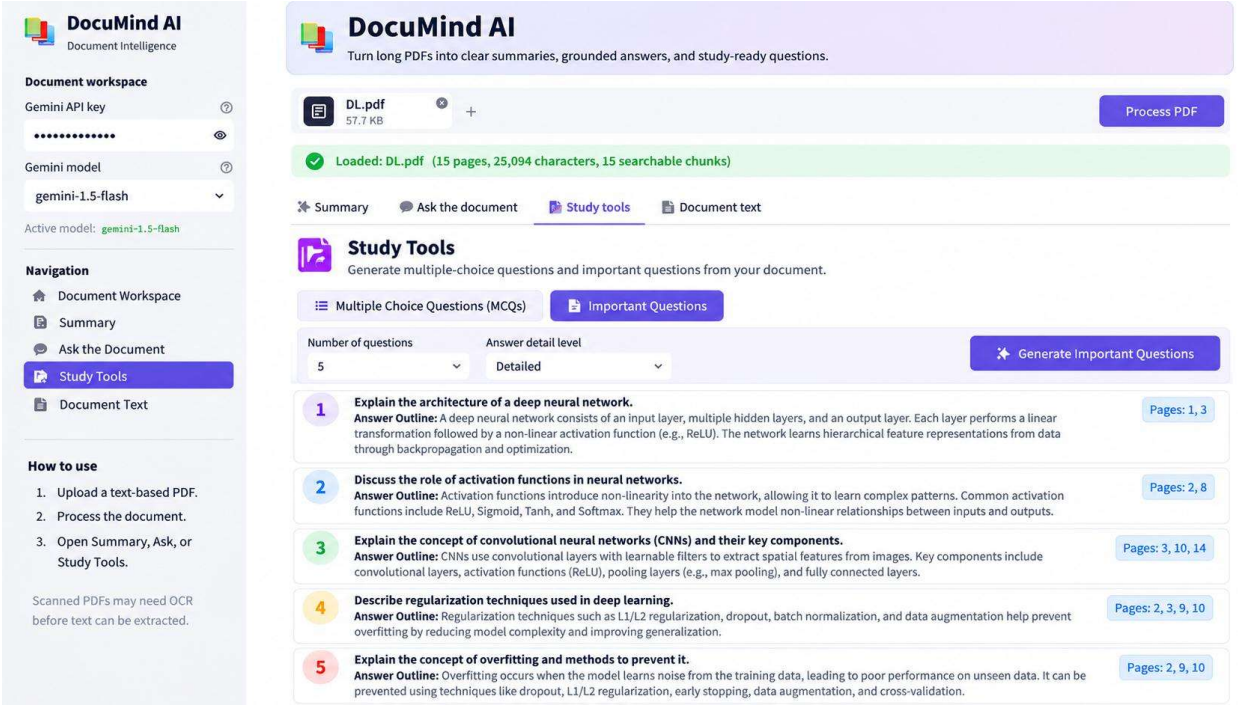
- Question-answering interface — Shows a user question, the generated answer, and supporting page references.



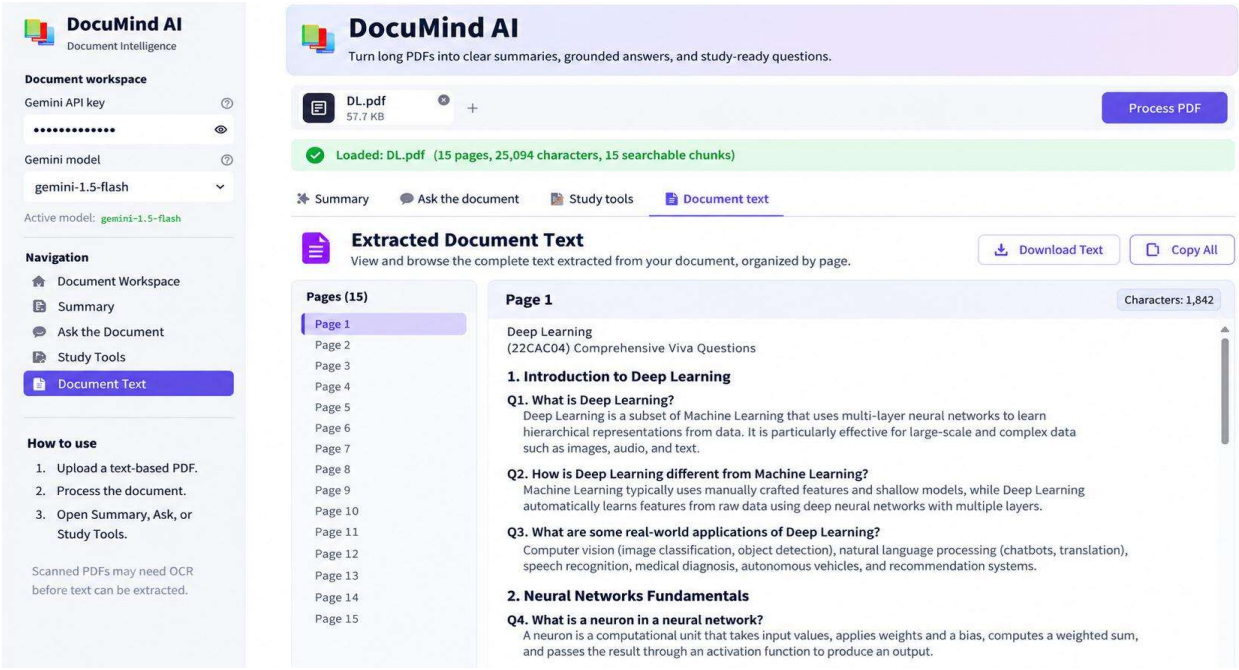
- MCQ generation — Displays multiple-choice questions, answer options, correct answers, and explanations.



- Important question generation — Produces exam-preparation questions with concise answer outlines.



- Extracted document text — Presents the extracted content organized by page.



Analysis — Bloom's L4

The experiments demonstrate that DocuMind AI combines PDF text extraction, TF-IDF-based information retrieval, and Large Language Model (LLM) capabilities to support document summarization and question answering. The PDF processing module extracts text while preserving page numbers, and the retrieval module identifies relevant passages to provide contextual information for generating responses.

The application successfully processed the displayed DL.pdf document, extracting text from 15 pages, creating 15 searchable chunks, and identifying 25,094 characters. The generated document overview, key points, and important terminology demonstrate how lengthy educational content can be organized into concise and structured study material.

The question-answering module generates responses based on retrieved document passages and provides supporting page references, allowing users to verify the information against the original document. The study tools also generate multiple-choice questions with answer explanations and important exam-preparation questions with answer outlines. These features demonstrate the system's ability to transform extracted document content into different forms of learning material.

However, the quality of the generated responses depends on the relevance of the retrieved passages and the language model's interpretation of the document. TF-IDF-based retrieval may miss semantically related passages when they use different terminology, while summarization of lengthy documents using representative passages may omit some details. Therefore, the observed results demonstrate successful execution of the implemented features, but do not establish a measured accuracy score or prove that every generated answer is correct. Further evaluation using manually verified questions, reference summaries, and citation checks would be required to assess reliability.

Evaluation — Bloom's L5

Strengths

- Automated document summarization: Generates concise document overviews, key points, and important terminology from uploaded PDFs.
- Document-based question answering: Uses TF-IDF-based retrieval and an LLM to answer questions using relevant document passages.
- Source traceability: Provides supporting page references to help users verify generated answers against the original document.
- Study material generation: Creates multiple-choice questions with answer explanations and important exam-preparation questions with answer outlines.
- Interactive dashboard: Provides a user-friendly interface for document upload, processing, analysis, and viewing extracted text.
- Downloadable outputs: Allows users to access and export generated summaries, study materials, and extracted document text.

Weaknesses

- The system primarily supports text-based PDFs; scanned documents may require OCR before text can be extracted.
- TF-IDF retrieval relies on lexical similarity and may fail to identify semantically relevant passages that use different words.
- Summarization of lengthy documents using representative passages may omit information from sections that are not selected.
- Generated answers and study questions may contain inaccuracies or unsupported information despite the use of page references.
- The observed results demonstrate successful execution of the displayed features, but systematic accuracy and reliability evaluations have not yet been performed.

Suggestions

Future improvements can include:

- Semantic search using embedding models for better contextual retrieval.
- Hierarchical summarization to preserve important information from lengthy documents.
- OCR integration for scanned PDFs and image-based documents.
- Improved citation validation to ensure that generated answers are supported by the referenced pages.
- Systematic evaluation using manually verified questions, reference summaries, and retrieval metrics.
- Caching and optimized processing to improve performance with larger documents.
- Support for additional file formats, such as DOCX and TXT.
- Enhanced answer verification and uncertainty handling to reduce unsupported responses.

Conclusion

The project demonstrated an AI-powered document summarization and question-answering system that combines PDF text extraction, TF-IDF-based information retrieval, prompt engineering, and a large language model. DocuMind AI successfully processed the tested document and generated summaries, key points, question-answering responses with page references, multiple-choice questions, and important exam-preparation questions. The interactive Streamlit dashboard also provides document navigation and downloadable outputs, making the system useful for educational and research applications. Further improvements in semantic retrieval, long-document summarization, citation verification, and systematic evaluation can enhance its reliability, scalability, and practical usefulness.